

JOTHIPRAKASH THANGARAJ

SOFTWARE ENGINEER

Backend Systems · Distributed Systems · Machine Learning · AI
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PROFESSIONAL SUMMARY

Software engineer with experience in software debugging, backend development, and systems engineering. Interested in building and understanding software from fundamentals, with hands-on work spanning Java, backend systems, distributed systems, system design, machine learning, and AI applications. Built and explored systems involving distributed file storage, media streaming, idempotent processing, in-memory data servers, retrieval-augmented generation, and GPT-style language models. Strong focus on learning through implementation and developing practical understanding of how software systems work.

TECHNICAL SKILLS

Languages	Java · Python · SQL · JavaScript · TypeScript
Backend & APIs	Spring Boot · WebFlux · REST APIs · WebSockets · FastAPI
Databases & Messaging	PostgreSQL · MySQL · Redis · Kafka · RabbitMQ
Distributed Systems	Caching · Rate Limiting · Fault Tolerance · Idempotency · Distributed Storage · Messaging
ML / AI	PyTorch · scikit-learn · NumPy · Pandas · MLflow · Hugging Face · RAG · LLMs
DevOps & Infrastructure	Docker · Kubernetes · Git · Linux · GitHub Actions

SELECTED ENGINEERING WORK

FileStore

Distributed remote file server backend focused on building storage infrastructure and handling files beyond a simple local filesystem.

github.com/TJothiprakash/filestore

StreamSpace

Streaming-focused system composed of a backend, UI, and FFmpeg worker, exploring components involved in serving and processing media.

github.com/TJothiprakash/streamspace-backend · [streamspace-ui](https://github.com/TJothiprakash/streamspace-ui) · [streamspace-ffmpeg_worker](https://github.com/TJothiprakash/streamspace-ffmpeg_worker)

Idempotency Processing

Backend engineering work exploring idempotent processing — designing operations so repeated requests can be handled safely.

github.com/TJothiprakash/idempotency-processing

Redis Server Lite

Lightweight Redis server implementation in Java, built to explore how an in-memory data server works from the inside.

github.com/TJothiprakash/redis_server_lite_java

RAG MLOps

ML engineering project exploring retrieval-augmented generation together with workflows for operationalizing ML applications.

github.com/TJothiprakash/rag-mlops

MiniGPT

Implementation-oriented project exploring the fundamentals behind GPT-style language models.

github.com/TJothiprakash/minigpt

PROFESSIONAL EXPERIENCE

Zoho Corporation — Software Debug Engineer

October 2023 – May 2024

- Investigated reported software problems by analyzing code and logs.
- Used debugging tools such as logcat, GDB, and IDE debuggers to identify root causes.
- Collaborated with development teams to implement fixes and verify effectiveness.
- Handled support tickets and worked to maintain SLA commitments.
- Identified database corruption issues and worked with PostgreSQL and MSSQL databases.
- Troubleshoot SMB, firewall, ports, and network connectivity issues.
- Documented debugging processes and maintained records of resolved issues.

Madura Coats — Manufacturing Executive

November 2018 – July 2021

- Planned operations to meet established schedules, factoring in order demands and business forecasts.
- Worked toward daily and monthly production targets while maintaining consistent quality.
- Analyzed quality issues and customer complaints and supported corrective and preventive actions.
- Maintained production logs and submitted timely operational reports.
- Supported 5S improvements and health and safety practices.
- Worked across cone winding, tube winding, hank winding, polishing, packing, spool winding, gassing, and reeling processes.

ENGINEERING FOUNDATIONS

DSA & Problem Solving

Data structures, algorithms, complexity analysis, and interview-oriented problem solving with extensive Java practice.

Java & Software Engineering

Java, OOP, abstraction, programming principles, design patterns, API development, and backend engineering.

System Design & Distributed Systems

Hands-on exploration of scalable systems, distributed patterns, caching, messaging, fault tolerance, storage, and backend architecture.

ML & AI Engineering

Machine learning fundamentals, RAG systems, LLM experimentation, AI applications, and agent-oriented engineering.

ADDITIONAL SELECTED WORK

Backend & Distributed Systems: Rate limiter · Circuit breaker · Bloom filter · Cache proxy · URL shortener · Broadcast server · WebSocket chat

AI / ML & MLOps: Input Pipeline MLOps · Article Recommender · Legal Aid Agent · AI Voice Assistant · Telco Churn MLOps

Foundations: Striver Sheet · Important Java Concepts · Programming Principles · Design Patterns · DSA practice

EDUCATION

K S R College of Technology, Tiruchengode

Bachelor of Technology in Textile Technology · 2014 – 2018 · CGPA 8.92

GBHSS Uthukuli, Tiruppur

Higher Secondary Certificate (HSC) · Bio-Maths · 2012 – 2014 · 87%

GHS Morattupalayam, Tiruppur

Secondary School Leaving Certificate (SSLC) · 2011 – 2012 · 86%

SELECTED GITHUB WORK

A broader collection of engineering work spans backend systems, distributed systems, Java, data structures and algorithms, AI/ML, MLOps, system design, and software engineering practice.

GitHub: github.com/TJothiprakash?tab=repositories